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ANA 101: Human Anatomy I - Cells, Tissues and the Skeletal System

Learn anatomy by systems and relationships rather than by isolated lists

FACULTY PRIMER

Faculty Primer

Faculty of Basic Medical Sciences
Human Anatomy | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

Anatomy has a reputation for impossible memorisation. It becomes manageable when you learn structures by relationship, function and clinical relevance. This pack covers anatomical terminology, tissue types, bone structure and the axial and appendicular skeleton, with a revision method built around drawing and recall.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Use correct anatomical position, planes, directional terms and movement terminology.
- Describe the four basic tissue types and identify their functions.
- Explain bone structure, growth and ossification, and classify bones by shape.
- Name and locate the major bones of the axial and appendicular skeleton.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

The language of anatomy

Precise terminology prevents dangerous ambiguity. Learn it before you learn structures.

- Anatomical position: standing upright, palms facing forward, feet together.
- Planes: sagittal (left and right), coronal (front and back), transverse (upper and lower).
- Directional terms: superior and inferior, anterior and posterior, medial and lateral, proximal and distal, superficial and deep.
- Movements: flexion, extension, abduction, adduction, rotation, circumduction, pronation and supination.
- Body cavities: cranial, vertebral, thoracic, abdominal and pelvic.

- Regions: use the correct regional term, for example brachial for arm and femoral for thigh.

> **Name the relationship, not just the part**
> In practicals you will be asked where a structure lies relative to another. Practise saying 'the median nerve passes medial to the brachial artery' rather than only labelling a diagram.

The four basic tissues

Organs are combinations of four tissue types, each specialised for a job.

- Epithelial tissue: covers and lines surfaces; classified by cell shape and layer number.
- Connective tissue: supports, binds and protects; includes bone, cartilage, adipose and blood.
- Muscle tissue: skeletal (voluntary, striated), cardiac (involuntary, intercalated discs), smooth (involuntary, non-striated).
- Nervous tissue: neurons transmit impulses; neuroglia support and protect.
- Histology requires practice at identifying tissue from appearance, not from written descriptions alone.
- Always relate structure to function: thin squamous epithelium suits diffusion; stratified epithelium suits protection.

Bone and the skeleton

Bone is living tissue that remodels throughout life, responding to load and hormonal signals.

- Long bone structure: diaphysis, epiphyses, periosteum, medullary cavity, articular cartilage.
- Compact bone forms the outer shell; spongy bone forms the inner lattice.
- Ossification: intramembranous for flat bones of the skull; endochondral for most long bones.
- Bone cells: osteoblasts build, osteocytes maintain and osteoclasts resorb.
- Axial skeleton: skull, vertebral column, ribs and sternum.
- Appendicular skeleton: pectoral girdle, upper limbs, pelvic girdle and lower limbs.

> **Draw from memory daily**
> Ten minutes of blank-page drawing beats an hour of highlighting. Draw, label, check, correct, repeat the next day.

The language of the body

Anatomy is a naming discipline before it is anything else. Terms of position - superior, inferior, anterior, posterior, medial, lateral, proximal, distal - locate a structure unambiguously regardless of posture. Planes (sagittal, coronal, transverse) allow the body to be described in sections. Without this vocabulary, description collapses into pointing.

The body is organised in levels: cells, tissues, organs, systems, organism. The four basic tissues - epithelial, connective, muscular and nervous - build every organ, and function follows from which tissues dominate and how they are arranged.

Anatomy is learned by region and by system at once. The regional approach answers what is in this part of the body; the systemic approach answers how a function works across it. Examinations test both, and strong answers move between them.

Skeleton, muscle and control

The skeleton supports, protects, stores minerals and produces blood cells. Bone features - foramina, fossae, condyles, tubercles - mark where structures pass, articulate or attach. Learning a bone means learning its landmarks, because that is what clinical examination and imaging rely on.

Skeletal muscle moves joints by contracting, working in antagonistic pairs. Origin, insertion, action and nerve supply are the four facts for every muscle, and the nerve supply is the one students most often omit. Lever systems explain how the same force produces different movement depending on where it attaches.

Control runs through the nervous and endocrine systems. The peripheral nervous system divides into somatic and autonomic, and the autonomic into sympathetic and parasympathetic divisions that generally oppose each other. Blood supply and lymphatic drainage explain how disease spreads and how drugs reach tissue.

> **Learn by layers**
> For any region: skin, fascia, muscle, vessel, nerve, bone. It is the order a surgeon dissects and the order an examiner expects.

Studying anatomy effectively

Anatomy is not learned by reading. It is learned by retrieval: labelling diagrams from memory, drawing structures from scratch, and explaining a region aloud without notes. Spaced repetition of those tasks over weeks outperforms rereading.

Use prosections, models and imaging with the respect the setting requires. Photograph and label your own diagrams rather than collecting images passively; constructing the label is the learning.

Clinical correlation is the strongest memory aid. A structure remembered because of what happens when it is damaged is retained far longer than one memorised from a list. Learn one clinical consequence for every major structure.

WORKED EXAMPLES

Do it once, slowly

These are not model answers to copy. They show the reasoning a strong student would use, step by step, so you can repeat the process on a question you have not seen.

1. Describing a region systematically

Describe the anterior compartment of the forearm.

1. Boundaries: fascia anteriorly, interosseous membrane posteriorly, elbow proximally, wrist distally.
2. Muscles: superficial and deep groups with their actions.
3. Nerve supply: median nerve, with ulnar contribution to one muscle.
4. Vessels: radial and ulnar arteries and branches.
5. Clinical correlation: the deficit following median nerve lesion at the wrist.
6. Present as a table rather than prose for revision.

> **What to carry forward**
> Boundaries, contents, supply, drainage, clinical correlation. Same order every time.

2. A functional anatomy answer

"Explain why the shoulder joint is prone to dislocation."

1. Classify: synovial, ball and socket, multiaxial.
2. Describe the bony arrangement: large head, shallow glenoid cavity.
3. Explain soft tissue contribution: labrum, capsule, ligaments, rotator cuff.
4. State the trade-off: mobility bought at the cost of stability.
5. Name the common direction of dislocation and why.
6. Conclude with the clinical implication.

> **What to carry forward**
> Structure, trade-off, consequence.

EXAM TECHNIQUE

Questions with model answers

Written in the shape your course is usually examined, with the marking logic spelled out. Attempt each one under time before you read the outline.

Q1. Name and describe the four basic tissue types with one location of each.

15 marks

1. Epithelial: covering and lining, avascular, e.g. epidermis.
2. Connective: support and binding, e.g. bone or tendon.
3. Muscular: contraction, e.g. skeletal muscle.
4. Nervous: conduction, e.g. spinal cord.
5. Link structure to function.

Q2. Explain the antagonistic arrangement of skeletal muscles with an example.

10 marks

1. Define agonist, antagonist and synergist.
2. Elbow flexion: biceps against triceps.
3. Explain why one muscle cannot reverse its own action.
4. Note the role of nerve supply in coordination.

REFERENCE

Glossary

Use these definitions in your own sentences; a copied definition rarely earns full marks.

Anatomical position. Standing upright, palms forward, the reference posture for directional terms.

Sagittal plane. A vertical plane dividing the body into left and right parts.

Proximal. Closer to the point of attachment or trunk.

Epithelium. Avascular tissue covering surfaces and lining cavities.

Origin. The relatively fixed attachment of a muscle.

Insertion. The attachment that moves during contraction.

Agonist. The muscle primarily responsible for a movement.

Antagonist. The muscle opposing the agonist.

Foramen. An opening in bone through which structures pass.

Synovial joint. A freely movable joint with a fluid-filled cavity.

Autonomic nervous system. The division controlling involuntary functions.

Clinical correlation. The functional consequence of a structure being damaged.

Your four-week plan

Week 1 - Terminology

Precision.

- Learn all directional terms.
- Label three planes.
- Describe one region aloud without notes.

Week 2 - Skeleton

Landmarks.

- Learn 30 bony landmarks.
- Identify what passes through three foramina.
- Draw the skull from memory.

Week 3 - Muscle

Action.

- Tabulate four facts for 20 muscles.
- Explain one lever system.
- Test a partner on actions.

Week 4 - Systems

Integration.

- Trace one nerve root to target.
- Trace blood supply to one limb.
- Write a functional anatomy answer.

Where to read more

Entry-level texts you can find in the university library or a departmental reading list. Confirm editions with your lecturer.

- Any standard regional anatomy textbook with clinical notes.
- An anatomy labelling workbook for active recall.
- A clinical anatomy text linking structures to common injuries.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. Define anatomical position.

Answer: Standing erect, face forward, upper limbs by the sides with palms facing forward, and feet together.

2. Which tissue type lines the inner surface of blood vessels?

Answer: Simple squamous epithelium, called endothelium, suited to diffusion and a smooth flow surface.

3. Distinguish intramembranous from endochondral ossification.

Answer: Intramembranous ossification forms bone directly from mesenchyme, as in skull flat bones. Endochondral ossification replaces a cartilage model, as in most long bones.

4. Name the bone cells responsible for bone resorption.

Answer: Osteoclasts.

5. Which plane divides the body into equal left and right halves?

Answer: The midsagittal plane.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Write the anatomical position and all directional terms from memory.
- [ ] Draw and label a long bone with six structures.
- [ ] Identify the four tissue types from microscope images.
- [ ] Name the bones of the axial skeleton without notes.
- [ ] Label the bones of the upper and lower limbs on a blank diagram.
- [ ] Explain the roles of osteoblasts, osteocytes and osteoclasts.
- [ ] Practise ten spotting questions with a study partner.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date: